Date: 09/04/2026

Duration: 3 hours

FINAL ASSESSMENT

II YEAR – SEMESTER 4

CSE23AE304 - PCP III

(B.Sc. E61)

SET H

Question 1: Problem Statement

You are given a string **S** consisting of lowercase English letters.

You may remove **any number of characters** from the string, but the **relative order of the remaining characters must remain the same**.

Your task is to construct the **lexicographically largest subsequence** possible from the string.

A string **A** is lexicographically larger than string **B** if at the first position where they differ, the character in **A** appears later in the alphabet than the character in **B**.

Return the resulting **largest subsequence**.

Example 1

Input

abacb

Output

ccb

Explanation

We pick characters greedily:

a b a c b

↑

Start from the largest possible characters moving forward.

The resulting subsequence is **ccb**.

Question 2: Problem Statement

You are given:

- An array tokens[] representing different token values
- An integer target representing the required value

You can use **any token multiple times**.

Your task is to find the **minimum number of tokens** required to reach exactly the target value.

If it is not possible, return -1.

Example 1

Input: tokens = [1,2,5], target = 11

Output: 3

Explanation:

5 + 5 + 1 = 11

Question 3: Problem Statement

You are given a grid of size m × n where:

- 0 represents an empty cell
- 1 represents an obstacle

A robot starts at the **top-left corner (0,0)** and wants to reach the **bottom-right corner (m-1, n-1)**.

The robot can only move:

- Right →
- Down ↓

Return the **number of unique paths** from start to end such that the robot **never steps on a blocked cell**.

Example 1

Input: grid = [

[0,0,0],

[0,1,0],

[0,0,0]

]

Output: 2

Explanation:

Two valid paths avoid the obstacle

Question 4: Problem Statement

You are working on a **corporate audit platform** that validates numeric console reports generated by legacy systems.

Since the system cannot compare visual console output directly, every report must be returned as a **collection of text rows**, where:

- Each row is stored as a **string**
- The entire report is returned as an **array/list of strings**
- The layout must remain **vertically and horizontally symmetric**
- Numbers represent **column positions**

Your task is to generate the expected report layout based on a given integer N.

Example 1

Input

3

Output

```
[
  "  1",
  " 121",
  "12321",
  " 121",
  "  1"
]
```

Question 5: Problem Statement

A logistics company needs to load packages onto a truck for delivery. The truck has a fixed **maximum weight capacity**.

You are given a list of packages:

packages = [[weight1, value1], [weight2, value2], ...]

Each package contains:

- **weight** → weight of the package
- **value** → value of the package

Unlike traditional problems, **packages can be partially loaded**. If only a fraction of a package is loaded, the value gained is proportional to the fraction of the weight taken.

For example:

- A package with weight = 10 and value = 100
- If 5 units are loaded → value gained = 50

Your task is to determine the **maximum total value** that can be loaded onto the truck without exceeding its capacity.

Example 1

Input

packages = [[10,60],[20,100],[30,120]]

capacity = 50

Output

240

Explanation

Steps:

- Take (10,60) fully → 60
- Take (20,100) fully → 100
- Remaining capacity = 20
- Take 20/30 of last → 80

Total = **240**